

Lecture Notes on Thermodynamic Computing

Chapter 7: Thermodynamic Processors and CPUs

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1 7.1 Introduction to Thermodynamic Processors

• Thermodynamic Processor Definition

Definition 1.1 (Thermodynamic Processor). A processing unit that operates at the thermodynamic energy scale, utilizing thermal fluctuations as computational resources while maintaining information processing capabilities at energies approaching $k_B T$ per operation.

Key characteristics:

- Ultra-low voltage operation ($V_{DD} < 5V_T$)
- Probabilistic instruction execution
- Thermal noise-assisted computation
- Energy per operation: $10 - 100 \times k_B T$
- Fault-tolerant architecture

• Architecture Overview

Thermodynamic processor components:

- Stochastic execution units
- Probabilistic memory hierarchy
- Thermal noise generators
- Adaptive voltage control
- Error correction systems

• Performance Metrics

Energy efficiency:

$$\eta = \frac{\text{Operations/second}}{\text{Power (Watts)}} = \frac{\text{OPS}}{\text{W}} \quad (1)$$

Thermodynamic efficiency:

$$\eta_{thermo} = \frac{E_{operation}}{k_B T} \geq 1 \quad (2)$$

Reliability metric:

$$R = P(\text{correct result}) \geq 1 - \epsilon_{target} \quad (3)$$

2 7.2 Instruction Set Architecture

• Probabilistic Instructions

Deterministic instructions:

$$\text{ADD R1, R2, R3: } R1 = R2 + R3 \quad (4)$$

Probabilistic instructions:

$$\text{PADD R1, R2, R3, p: } R1 = \begin{cases} R2 + R3 & \text{with probability } p \\ \text{undefined} & \text{with probability } 1 - p \end{cases} \quad (5)$$

Stochastic operations:

$$\text{SMUL R1, R2, R3: } R1 = R2 \times R3 \text{ (stochastic)} \quad (6)$$

• Error-Aware Instructions

Instructions with built-in error detection:

$$\text{SADD R1, R2, R3, E: } R1 = R2 + R3, E = \text{error_flag} \quad (7)$$

Approximate computing instructions:

$$\text{AADD R1, R2, R3, } \epsilon : \text{AADD R1, R2, R3, } \epsilon : \text{AADD R1, R2, R3, } \epsilon : \text{AADD R1, R2, R3, } \epsilon : |R1 - (R2 + R3)| \leq \epsilon \quad (8)$$

• Thermal Control Instructions

Temperature adjustment:

$$\text{SETTEMP T: } T_{\text{effective}} = T \quad (9)$$

Noise level control:

$$\text{SETNOISE } \sigma : \text{SETNOISE } \sigma : \text{SETNOISE } \sigma : \text{SETNOISE } \sigma : \sigma_{\text{thermal}} = \sigma \quad (10)$$

Energy budget instruction:

$$\text{SETBUDGET E: } E_{\text{max}} = E \times k_B T \quad (11)$$

3 7.3 Execution Units

• Stochastic Arithmetic Logic Unit

ALU architecture:

Operation mapping:

$$\text{ADD} \rightarrow \text{Scaled OR gates} \quad (12)$$

$$\text{MUL} \rightarrow \text{AND gates} \quad (13)$$

$$\text{SQRT} \rightarrow \text{JK flip-flops} \quad (14)$$

$$\text{SIG} \rightarrow \text{Comparator chains} \quad (15)$$

Energy per operation:

$$E_{\text{ALU}} = N_{\text{gates}} \times C_{\text{gate}} \times V_{DD}^2 \times \alpha \quad (16)$$

where α is switching activity factor.

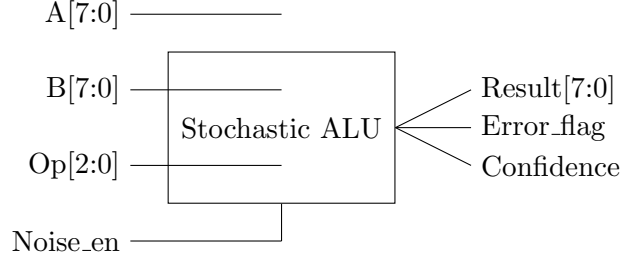


Figure 1: Stochastic ALU block diagram

• Floating Point Unit (FPU)

Thermodynamic FPU challenges:

- Mantissa precision vs energy trade-off
- Exponent overflow in low voltage
- Rounding error accumulation

Approximate floating point format:

$$\text{Value} = (-1)^s \times (1 + M \pm \epsilon_M) \times 2^{E \pm \epsilon_E} \quad (17)$$

Error propagation:

$$\epsilon_{result} = \sqrt{\epsilon_A^2 + \epsilon_B^2 + \epsilon_{operation}^2} \quad (18)$$

• Vector Processing Unit

SIMD operations with thermal noise:

$$\mathbf{C} = \mathbf{A} \odot \mathbf{B} + \boldsymbol{\eta} \quad (19)$$

where $\boldsymbol{\eta}$ is thermal noise vector.

Correlation effects:

$$\text{Corr}(C_i, C_j) = \text{Corr}(A_i, A_j) \times \text{Corr}(B_i, B_j) + \delta_{ij} \sigma_{thermal}^2 \quad (20)$$

4 7.4 Memory Hierarchy

• Thermodynamic Cache Design

Cache energy model:

$$E_{cache} = E_{tag} + E_{data} + E_{thermal} \quad (21)$$

Tag comparison in subthreshold:

$$P_{match} = \prod_{i=0}^{n-1} \left[1 - \frac{1}{2} \exp \left(-\frac{(V_{tag,i} - V_{addr,i})^2}{2\sigma_{noise}^2} \right) \right] \quad (22)$$

- **kT-RAM Integration**

Cache line organization:

$$\text{Line} = \{\text{Valid}, \text{Tag}, \text{Data}, \text{ECC}, \text{Confidence}\} \quad (23)$$

Replacement policy with uncertainty:

$$\text{LRU_score} = \text{Age} \times (1 + \epsilon_{\text{uncertainty}}) \quad (24)$$

Thermal refresh requirements:

$$f_{\text{refresh}} = \frac{1}{\tau_{\text{retention}}} \times \frac{\ln(BER_{\text{target}})}{\ln(1 - P_{\text{error}})} \quad (25)$$

- **Approximate Memory**

Multi-level error tolerance:

$$\text{Memory} = \{\text{Critical}, \text{Important}, \text{Approximate}\} \quad (26)$$

Energy allocation:

$$E_{\text{critical}} : E_{\text{important}} : E_{\text{approximate}} = 10 : 5 : 1 \quad (27)$$

5 7.5 Pipeline Architecture

- **Probabilistic Pipeline**

Pipeline stages:

1. Fetch (with thermal bit errors)
2. Decode (approximate instruction parsing)
3. Execute (stochastic computation)
4. Memory (kT-RAM access)
5. Writeback (error correction)

Throughput with errors:

$$\text{IPC} = \text{IPC}_{\text{ideal}} \times \prod_{i=1}^5 (1 - P_{\text{error},i}) \quad (28)$$

- **Error Propagation Model**

Stage error accumulation:

$$P_{\text{total}} = 1 - \prod_{i=1}^N (1 - P_{\text{stage},i}) \quad (29)$$

Error masking probability:

$$P_{\text{mask}} = P(\text{error doesn't affect output}) \quad (30)$$

Pipeline recovery:

$$\text{Recovery_cycles} = \frac{\text{Detection_latency}}{P_{\text{detection}}} \quad (31)$$

- **Dynamic Voltage Scaling**

Per-stage voltage control:

$$V_{stage}(t) = V_{min} + (V_{max} - V_{min}) \times \text{Load_factor}(t) \quad (32)$$

Power-performance trade-off:

$$f_{max} = \frac{I_{max}}{C_{load}V_{DD}} \propto \exp\left(\frac{V_{DD}}{nV_T}\right) \quad (33)$$

Energy optimization:

$$E_{pipeline} = \sum_{i=1}^N \frac{1}{2} C_i V_i^2 + V_i I_{leak,i} t_{stage,i} \quad (34)$$

6 7.6 Fault Tolerance and Reliability

- **Error Detection and Correction**

Hamming codes for probabilistic data:

$$\mathbf{r} = \mathbf{d} + \mathbf{e} + \boldsymbol{\eta} \quad (35)$$

where $\mathbf{e}$ is error vector and $\boldsymbol{\eta}$ is thermal noise.

Syndrome calculation:

$$\mathbf{s} = \mathbf{H}\mathbf{r}^T = \mathbf{H}(\mathbf{e} + \boldsymbol{\eta})^T \quad (36)$$

Correction threshold:

$$\text{Correct if } \|\mathbf{s}\| > \theta_{thermal} \quad (37)$$

- **Redundant Execution**

Dual modular redundancy (DMR):

$$P_{undetected} = P_{error}^2 \quad (38)$$

Triple modular redundancy (TMR):

$$P_{failure} = 3P_{error}^2 - 2P_{error}^3 \quad (39)$$

Adaptive redundancy:

$$R(t) = \begin{cases} 1 & \text{if } P_{error} < \theta_{low} \\ 2 & \text{if } \theta_{low} \leq P_{error} < \theta_{high} \\ 3 & \text{if } P_{error} \geq \theta_{high} \end{cases} \quad (40)$$

- **Checkpoint and Recovery**

Checkpoint interval optimization:

$$T_{opt} = \sqrt{\frac{2C}{\lambda}} \quad (41)$$

where C is checkpoint cost and λ is failure rate.

Recovery energy:

$$E_{recovery} = E_{checkpoint} + E_{rollback} + E_{re-execution} \quad (42)$$

7 7.7 Performance Analysis

• Instruction Throughput

Instructions per cycle with errors:

$$IPC = \frac{N_{issued}}{N_{cycles}} \times P_{correct_execution} \quad (43)$$

Thermal noise impact:

$$P_{correct} = \exp\left(-\frac{N_{operations} \times P_{bit_error}}{P_{masking}}\right) \quad (44)$$

• Energy Efficiency Analysis

Performance per watt:

$$\frac{MIPS}{Watt} = \frac{f_{clock} \times IPC}{P_{total}} \quad (45)$$

Thermodynamic efficiency:

$$\eta_{thermo} = \frac{P_{useful_computation}}{P_{total}} \times \frac{k_B T}{E_{per_operation}} \quad (46)$$

• Scaling Laws

Technology scaling effects:

$$\frac{V_{DD}}{V_T} = \text{constant} \Rightarrow V_{DD} \propto T \quad (47)$$

Performance scaling:

$$\text{Performance} \propto \frac{1}{\text{Feature size}} \times \sqrt{T} \quad (48)$$

Power density:

$$\text{Power density} \propto \frac{f \times C \times V_{DD}^2}{\text{Area}} \propto \frac{T^{2.5}}{\text{Feature size}^2} \quad (49)$$

8 7.8 Compiler Support

• Error-Aware Compilation

Code transformation for fault tolerance:

$$\text{Original: } x = a + b \quad (50)$$

$$\text{Transformed: } x = \text{FADD}(a, b, \text{confidence_level}) \quad (51)$$

Loop unrolling with error checking:

$$\text{for } i = 0 \text{ to } N \text{ step } k \{ \quad (52)$$

$$\quad \text{Compute block of } k \text{ iterations} \quad (53)$$

$$\quad \text{if (error_detected) retry_block()} \quad (54)$$

$$\quad \} \quad (55)$$

• Approximate Computing Optimization

Quality-energy trade-off:

$$\text{Objective} = \alpha \times \text{Energy} + \beta \times \text{Error} \quad (56)$$

Precision allocation:

$$\text{Precision}[i] = \text{argmin}_p (\text{Energy}(p) + \lambda \times \text{Error}(p)) \quad (57)$$

• Thermal-Aware Scheduling

Temperature-dependent instruction scheduling:

$$\text{Schedule_time} = \text{Base_time} \times \left(1 + \alpha \times \frac{T - T_{ref}}{T_{ref}} \right) \quad (58)$$

Energy budget management:

$$\sum_{i=1}^N E_i \leq E_{budget} \times \text{Performance_requirement} \quad (59)$$

9 Examples

Example 9.1 (Thermodynamic Processor Energy Budget). Calculate energy budget for 1 GOPS thermodynamic processor at 300K.

Given:

- Target performance: 1 GOPS (billion operations per second)
- Energy per operation: $50 \times k_B T$
- Operating temperature: $T = 300$ K
- Overhead factor: 2.0 (including memory, control, etc.)

Solution:

Energy per operation:

$$E_{op} = 50 \times k_B T \quad (60)$$

$$= 50 \times 1.38 \times 10^{-23} \times 300 \quad (61)$$

$$= 2.07 \times 10^{-19} \text{ J} = 207 \text{ aJ} \quad (62)$$

Computation power:

$$P_{compute} = 10^9 \times 207 \times 10^{-18} \quad (63)$$

$$= 0.207 \text{ mW} \quad (64)$$

Total power with overhead:

$$P_{total} = 2.0 \times 0.207 = 0.414 \text{ mW} \quad (65)$$

Energy efficiency:

$$\eta = \frac{10^9}{0.414 \times 10^{-3}} = 2.4 \times 10^{12} \text{ OPS/W} \quad (66)$$

This is 10^4 better than conventional processors!

Example 9.2 (Pipeline Error Analysis). Analyze error propagation in 5-stage thermodynamic pipeline.

Given:

- 5 pipeline stages
- Error probability per stage: $P_e = 0.01$
- Error masking probability: $P_m = 0.3$
- Instructions per cycle: $IPC = 1.0$ (ideal)

Solution:

Total error probability:

$$P_{total} = 1 - (1 - P_e)^5 \quad (67)$$

$$= 1 - (0.99)^5 = 0.049 \quad (68)$$

Effective error probability with masking:

$$P_{effective} = P_{total} \times (1 - P_m) \quad (69)$$

$$= 0.049 \times 0.7 = 0.034 \quad (70)$$

Actual IPC:

$$IPC_{actual} = IPC_{ideal} \times (1 - P_{effective}) \quad (71)$$

$$= 1.0 \times (1 - 0.034) = 0.966 \quad (72)$$

Performance degradation: 3.4

10 Homework Problems

Assignment 7.1: Thermodynamic Processors and CPUs

1. Instruction Set Design:

- Design probabilistic instruction encoding
- Calculate instruction energy breakdown
- Implement error-aware execution model

2. ALU Architecture:

- Design 8-bit stochastic ALU
- Implement approximate floating point unit
- Optimize for energy-accuracy trade-off

3. Memory Hierarchy:

- Design thermodynamic cache system
- Integrate kT-RAM with error correction

- Optimize cache replacement policy
4. **Pipeline Design:**
 - Implement 5-stage thermodynamic pipeline
 - Design dynamic voltage scaling
 - Calculate optimal pipeline depth
 5. **Error Analysis:**
 - Model fault propagation through processor
 - Design adaptive redundancy system
 - Calculate reliability vs energy trade-off
 6. **Performance Modeling:**
 - Develop thermodynamic processor simulator
 - Analyze scaling laws with technology
 - Compare with conventional processors
 7. **Compiler Support:**
 - Design error-aware compiler passes
 - Implement approximate computing optimization
 - Develop thermal-aware code generation
 8. **System Integration:**
 - Design complete thermodynamic system
 - Implement adaptive control algorithms
 - Optimize for application-specific workloads

Next Chapter Preview: Chapter 8 will examine "Applications and Future Directions" including real-world applications and research frontiers in thermodynamic computing.